Higher National Diploma in Information Technology
Second Year, Second Semester Examination – 2020
HNDIT2413-Web Application Development

Instructions for Candidates:
Answer five (5) questions only.
All questions carry equal marks

No. of questions : 06
No. of pages :09
Time : Three (03) hours

01.

i. What do the following acronyms stand for?
   a. HTTP
   b. W3C [02 Marks]

ii. Write the three main functions of client server architecture. [03 Marks]

iii. Explain the following concepts with examples.
   a. Web Browser
   b. Search engine [04 Marks]

iv. Write HTML codes to create the following. [05 Marks]

Our Menus

1. Fried Rice
   - Sea food fried rice
   - Chicken fried rice
2. Pizza
   - Pan and other crust
     - A. Classic pizza
     - B. Supreme pizza
   - Thin crust

v. Write the output of the following HTML code. [06 Marks]

```
<table border="2" align=Center>
<TR><TH colspan=3>Resturant food for you</TH></TR>
        <TR>
                <TH>Item code</TH>
                <TH>Item</TH>
                <TH>Price</TH> </TR>
<TR>
                <TD>pizza-03 </TD>
                <TD>cheese pizza</TD>
                <TD>Rs.1000</TD></TR>
        <TR>
                <TD>Pizza-06 </TD>
                <TD>Spicy Seafood pizza</TD>
                <TD>Rs.2500</TD></TR>
<TR>
        <TD colspan=2 align=center>Total</TD>
        <TD>Rs.3500</TD>
</TR> </table>
```

[Total 20 Marks]

02.

i. What are the benefits of using CSS for a web site? [02 Marks]

ii. State three advantages of keeping CSS code in an external file. [03 Marks]

iii. Mention three ways to insert styles into web page. [03 Marks]

iv. Explain the following selectors in CSS with an example.

   a. CSS id Selector

   b. CSS Universal Selector

[04 Marks]

vi. By considering the following web page complete the following CSS style definitions by filling the missing code segments at the given placeholders to get the following appearance of the HTML page.

[08 Marks]

Food for you

Enjoy with our new Pizza!!!

category 01

Cheese Pizza

Cheese Pizza is a catergory of pizza rich with Rich tomato sauce with a triple layer of mozzarella cheese.

category 02

Tandoori Chicken Pizza

Tandoori Chicken Pizza is category of pizza with rich Tandoori chicken & onions with a double layer of mozzarella cheese

category 03

Veggie Supreme Pizza

Veggie Supreme Pizza is category of pizza with rich with Mushrooms, tomatoes, onions, black olives and bell peppers with a double layer of mozzarella cheese.

```
<html>
<head><link rel="stylesheet" href="style.css"></head>
        <body>
                <h1>Food for you</h1>
                <h2>Enjoy with our new Pizza!!!</h2>
                <hr>
        <div class="cat1">
        <h3>category 01</h3>
        <h3>Cheese Pizza</h3>
        <p>Cheese Pizza is a category of pizza rich with Rich
        tomato sauce with a triple layer of mozzarella cheese.</p>
        </div>
        <div class="cat2">
        <h3>category 02</h3>
        <h3>Tandoori Chicken Pizza</h3>
        <p>Tandoori Chicken Pizza is category of pizza with rich
        Tandoori chicken & onions with a double layer of mozzarella cheese.</p>
        </div>
        <div class="cat3">
        <h3>category 03</h3>
        <h3>Veggie Supreme Pizza</h3>
        <p>Veggie Supreme Pizza is category of pizza with rich with Mushrooms,
        tomatoes, onions, black olives and bell peppers with a double layer of mozzarella
        cheese.</p>
        </div>
        </body>
        </html>
```

```
body{
  (1)……..//Set full width of display fit to the web page
  (2)…… //Set margin 0px
  (3)…… //set center alignment for body text
}

      h1{}
      (4)…… //set font size 30px

      (5)…… //set horizontal line height to 10px
      div
      {
      padding-bottom:30px.//set bottom padding line 30px
      (6)……..//set width of div section to 30.5% from whole display
      float:left;
      position:relative;
      min-height:1px;
      (7)……..// set 5px margin in left side of div section
      }
      .cat1
      {
      (8)…… //set background color of div section(cat1)to #d5f4e6
      border-color:#bce8f1 // set border color of div section(cat1)to # bce8f1
      }
      cat2{}
      cat3{}
```

[Total 20 Marks]

03.

i. Mention two important features of Java script. [02 Marks]

ii. Give two examples for client-side scripting language. [02 Marks]

iii. What is HTML DOM? [02 Marks]

iv. Write the outputs of the following java script codes.

a. [02 Marks]

```
var a=3;
var b=a++;
var c=++a;
document.write(a,b,c);
```

b.

[02 Marks]

```
var x = 6;
var y = 3;

document.write(
!(x === y) + "<br>" +
!(x > y));
```

True
False

c.

[02 Marks]

```
var arr = [1, 2, 5, 10, 20];

var sum = 0;

for (var i in arr) {

 sum += arr[i];

}

document.write(sum);
```

d.

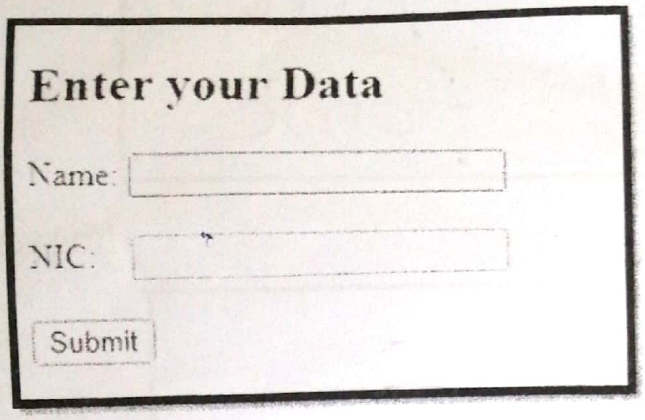
[02 Marks]

```
function myFunction(a, b) {

 return a * b;

}

var x = myFunction(4, 3);

document.write(x);
```

v. Considering the following HTML form write one Java script function for following data validations.

Name and NIC fields are not empty.

NIC field must be Nine (09) digits numeric number. [06 Marks]




```
<h2>Enter your Data</h2>
<form name="myForm"
action="/action_page.php"
onsubmit="return validateForm()"
method="post">
  Name: <input type="text" name="fname">
  <br><br>
  NIC:    <input
type="text" name="fname">
  <br><br>
  <input type="submit" value="Submit">
</form>
```

[Total 20 Marks]

04.

i. Briefly explain the server-side script technology and write two examples. [04 Marks]

ii. What are the three components need to install on the computer for developing and executing a php programme? [03 Marks]

iii. Write the outputs of the following PHP codes.

a. [02 Marks]

```
<?php
$x = 100;
$y = "100";
if ($x === $y) {
echo ("Numbers are identical");
}
else
{
echo ("Numbers are not identical");
}
?>
```

b. [02 marks]

```
<?php
$colors = array("Red", "Green", "Blue");
foreach ($colors as $value) {
  echo "$value[0] <br>";
}
?>
```

c.

[03 Marks]

```
<?php
Function add_sub($val1,$val2)
{
        $sum = $val1 + $val2;
        $sub=$val1 - $val2;
        return array($sum,$sub);
}
list($a,$b) = add_sub(5,4);
echo $a."<br>";
echo $b;
?>
```

iv. A Prime number is a natural number greater than 1 and divisible by 1 and itself only, for example: 2, 3, 5, 7, etc.

Write a PHP function called "primeCheck()" to check the entered number is prime number or not.

(Hint- by dividing number(N) with all natural numbers starting from 2 to N/2.)

[06 Marks]

[Total 20 Marks]

05.

i. Define PHP cookie. [02 Marks]

ii. PHP is object-oriented language. Declare a class named Fruit consisting of two properties ($name and $color) and two methods set_name () and get_name () for setting and getting the $name property.

[04 Marks]

iii. By considering the following PHP code as a multi-dimensional array and complete the blanks of the following (a) to (e) [05 Marks]

```
<?php
    $marks = ….(a)……(
      "Hasith" => …..(b)….. (
        "Web Development" => 85,
        "Mobile Application Development" => 80,)
    );
    echo "Marks for Hasith in Web development is : " ;
    echo …..(c)…….['…..(d)…..']['…….(e)……'] . "<br />";
?>
Out put:
        Marks for Hasith in web Development is : 85
```

iv. Considering the following HTML form write a PHP code to do the following tasks.

PHP Form -Student Information

* required field

Name : *

Age : *

Gender: ◯ Female ◯ Male ◯ Other *

Submit

a. All fields are required. (Hint-If name is empty generate a message as "Name is required") [06 Marks]

b. Once student enter correct data, display output as follows.

Your Input:
Sahan
20
Male

[03 Marks]

[Total 20 Marks]

06.

This form is used to store Employee Registration data in the MYSQL database server using PHP.

Employee Registration

Emp_No :

Name :

Gender :

Address :

Submit

Database server: localhost

User Name: root

Password: emp123

Database Name: Employee_DB

Table Name: Employee

i. Write HTML code to create the above form. [04 Marks]

ii. Write PHP code segment to create the connection and check the connection to the MYSQL database. [02 Marks]

iii. Write PHP code segment to select the database "Employee_DB" and create Employee table. [03 Marks]

iv. Write PHP code segment to insert a record into Employee table. [03 Marks]

v. Write PHP code segment to change the name to 'Amal' whose employee id is '1003'. [03 Marks]

vi. Write PHP code segment to display the records of Employee's whose gender is male. [05 Marks]

[Total 20 Marks]